Peter has 6 grey socks, six blue socks and four black socks in a drawer.The socks are not made up into pairs. Just as he picks a black sock there is a power failure. What is the probability that:

a. He will pull out a second black sock in the dark? $\frac{3}{15}$

b. Will end up with a pair of socks that are different in colour? $\frac{12}{15}$

3. Two dice are thrown and a table is made representing the possible **sum of the scores** on the two dice

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

a. Complete the table by filling in the sum of the scores on the dice.

b. What is the probability of getting a combined score of 15? $\frac{0}{36}$

c. What is the probability of getting a combined score of 7 or more? $\frac{21}{36}$

d. What is the probability of getting a combined score of 3 or less? $\frac{3}{36}$

MAS page 295 Ex 1

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The jokers are removed from a pack of playing cards, and it is then shuffled. Two cards are then drawn from the pack. Calculate the probability of drawing:

a. A queen on the first draw. $\frac{4}{52}$

b. An ace on the second draw, if:

i. The first card is replaced. $\frac{4}{52}$

ii. The first card is not an ace, and is not replaced. $\frac{4}{51}$

c. An ace and then a jack, if:

i. The first card is replaced. $\frac{4}{52} \times \frac{4}{52} = \frac{1}{169}$

ii. The first card is not replaced. $\frac{4}{52} \times \frac{4}{51} = \frac{4}{663}$

d. Two aces without the first card being replaced. $\frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$

5. An old lady keeps her money in a biscuit tin. She has three R100 notes, four R50 notes, seven R20 notes and six R10 notes. Determine the:

a. Probability that she draws two R100 notes. $\frac{3}{20} \times \frac{2}{19} = \frac{3}{190}$

b. Probability that the total is less than R100. $\frac{181}{190}$ ie Total − (HH + FF)

(Assume that she has drawn one note and then another).

More Probability terminology:

1. **SAMPLE SPACE:** The sample space of an activity is the set of all possible outcomes.

Example: A die is thrown and results are analysed.

List a set representing the sample space S = { 1 ; 2 ; 3 ; 4 ; 5 ; 6 }

$n(S)$ = 6

2. **EVENT:** In the die example above,

List the set representing all even outcomes E = { 2 ; 4 ; 6 }

$n(E)$ = 3

List the set containing all odd outcomes O = { 1 ; 3 ; 5 }

$n(O)$ = 3

What is the probability of throwing an even number?

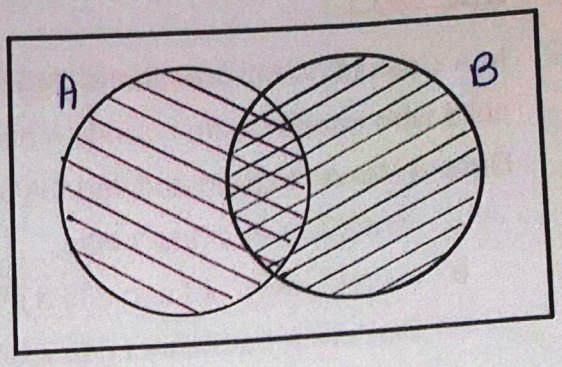
$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

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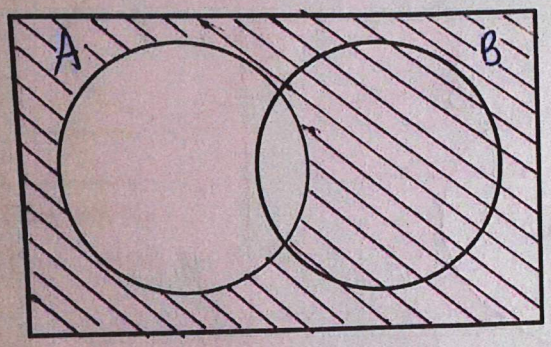
DUCTION TO VENN DIAGRAMS

a) The Sample space consists of set A and set B. Label the sets.
b) Shade the Area representing all things in set A
c) Shade differently the area representing all things in Set B
d) What does the overlapping area represent?

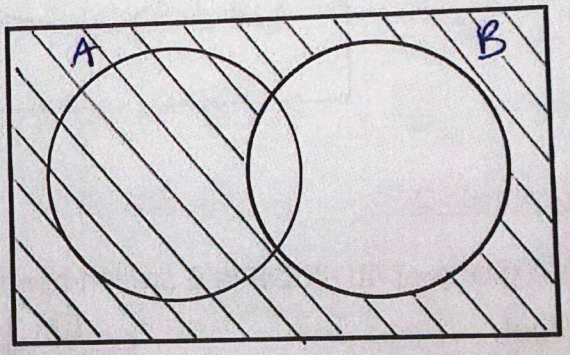
$A \cap B$



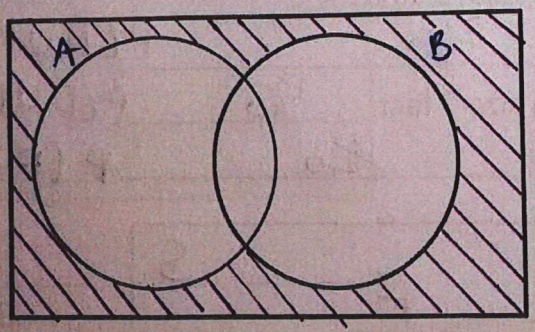
2. Shade the area indicating all things not in set A



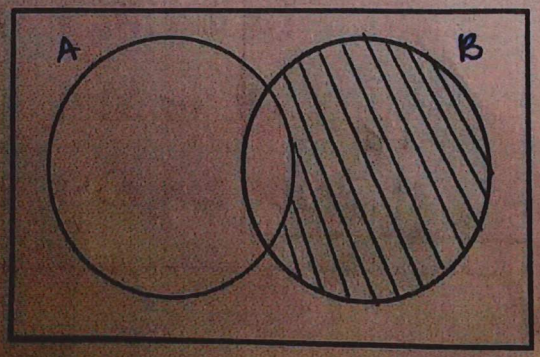
3. Shade the area indicating all things not in set B



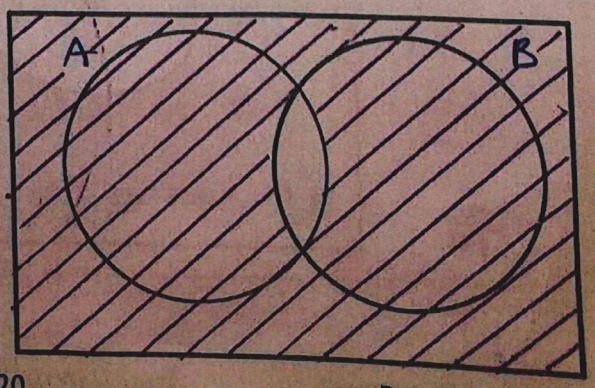
4. Shade the area representing all things not in A nor B



5. Shade the area representing all things
   i) not in A and in B



   ii) not in (A and B)



ise

The Venn diagrams represent the ways the events A and B can take place.
For each case, find:

a. $P(A \text{ and } B)$ b. $P(A \text{ but not } B)$ c. $P(A \text{ or } B)$ d. $P(B \text{ but not } A)$

Diagram (i)

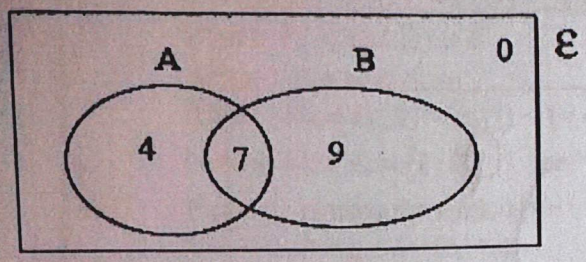


Diagram (i)

a. $P(A \text{ and } B) = \frac{7}{20}$

b $P(A, \text{not } B) = \frac{4}{20}$

c. $P(A \text{ or } B) = \frac{20}{20}$

d. $P(B, \text{not } A) = \frac{9}{20}$

Diagram (ii)

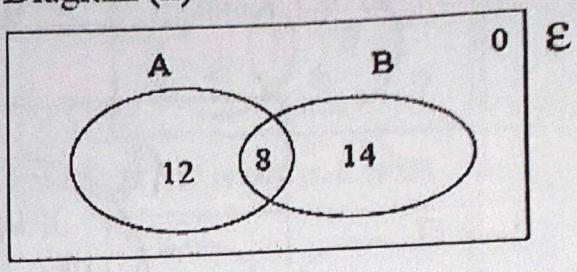


Diagram (ii)

a. $P(A \text{ and } B) = \frac{8}{34}$

b. $P(A, \text{not } B) = \frac{12}{34}$

c. $P(A \text{ or } B) = \frac{34}{34}$

d. $P(B, \text{not } A) = \frac{14}{34}$

Diagram (iii)

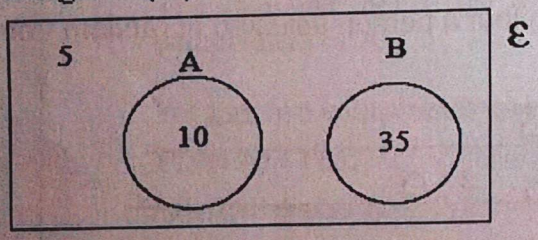


Diagram (iii)

a. $P(A \text{ and } B) = \frac{0}{50}$

b. $P(A, \text{not } B) = \frac{10}{50}$

c. $P(A \text{ or } B) = \frac{45}{50}$

d. $P(B, \text{not } A) = \frac{35}{50}$

Diagram (iv)

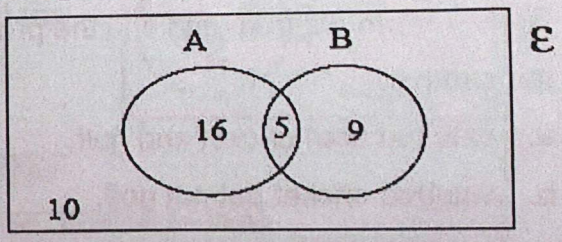


Diagram (iv)

a. $P(A \text{ and } B) = \frac{5}{40}$

b. $P(A, \text{not } B) = \frac{16}{40}$

c. $P(A \text{ or } B) = \frac{30}{40}$

d $P(B, \text{not } A) = \frac{9}{40}$

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Example 1

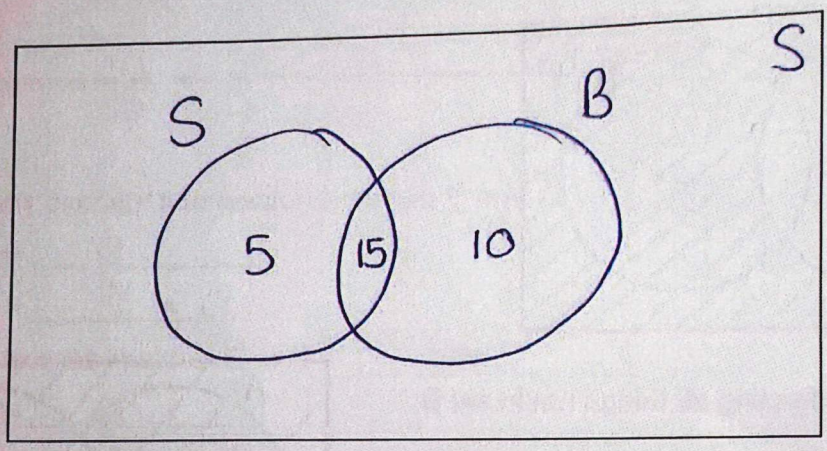
In a class of 30 children there are 20 who can swim and 25 who can ride a bike. There is no child who cannot swim nor ride a bike.

Draw a Venn diagram and find the probability that a child selected at random:

a. can swim and ride a bike. $\frac{15}{30}$ $P(S \cap B)$

b. cannot swim but can ride a bike. $\frac{10}{30}$ $P(B \cap S')$

c. can swim but cannot ride a bike. $\frac{5}{30}$ $P(S \cap B')$



Example 2

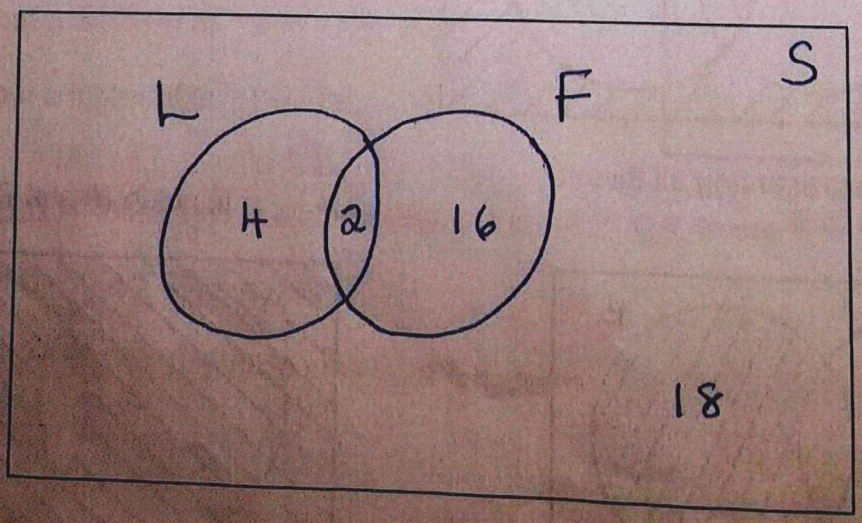
In a group of 40 students 6 are left-handed, 18 have size 7 feet and 2 are left-handed with size 7 feet.

Draw a Venn diagram and find the probability that a student is:

a. left-handed or has size 7 feet. $\frac{22}{40}$ $P(L \cup F)$

b. not left-handed and does not have size 7 feet. $\frac{18}{40}$ $P(L \cup F)'$

c. not left-handed and has size 7 feet. $\frac{16}{40}$ $P(F \cap L')$



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2. When a card is selected at random from a full pack of cards, two possible events are:

R: A red card is selected.

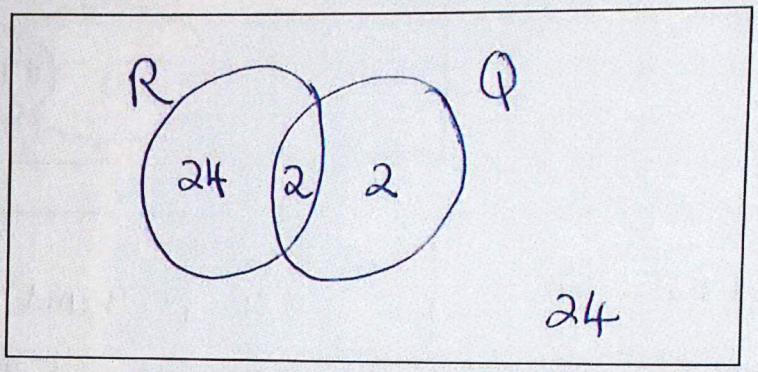
Q: A queen is selected.

Draw a Venn diagram and use it to find the probabilities that a card is:

a. a red queen. $\frac{2}{52}$

b. red or a queen $\frac{28}{52}$

c. not red or a queen. $\frac{28}{52}$



5. On a survey 100 people were asked whether they watched golf or cricket on TV.

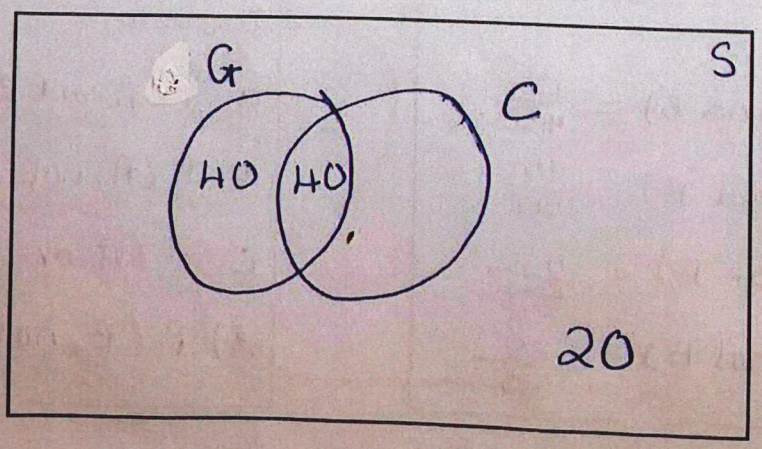
20 watched neither 80 watched golf 40 watched cricket.

Draw a Venn diagram and find the probabilities that a person selected at random from the sample:

a. watched both cricket and golf. $\frac{40}{100}$

b. watched cricket but not golf. $\frac{0}{100} = 0$

c. watched golf but not cricket. $\frac{40}{100}$



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